



Approximation algorithm for (connected) Italian dominating function

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ABSTRACT

For a connected graph $G = (V, E)$, a function $r_i : V \mapsto \{0, 1, 2\}$ is an Italian dominating function (IDF) if every vertex v with $r_i(v) = 0$ is either adjacent to a vertex u with $r_i(u) = 2$ or adjacent to at least two vertices x, y with $r_i(x) = r_i(y) = 1$. The weight of r_i is $\sum_{v \in V} r_i(v)$ and the minimum Italian dominating function problem (MinIDF) is to compute an IDF with minimum weight. The minimum connected Italian dominating function problem (MinCIDF) is to find a minimum weight IDF r_{ci} such that the subgraph of G induced by vertex set $\{v \in V \mid r_{ci}(v) = 1 \text{ or } r_{ci}(v) = 2\}$ is connected. In this paper, we give a greedy algorithm for the MinIDF problem with approximation ratio at most $2 + \ln(\frac{1}{2}\Delta + 1)$, and a greedy algorithm for the MinCIDF problem with approximation ratio at most $4 + \ln(\frac{1}{2}\Delta - 1)$, where Δ is the maximum degree of the graph.

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1. Introduction

In this paper, we study approximation algorithms for the minimum (connected) Italian dominating function problem. A function $r_i : V(G) \rightarrow \{0, 1, 2\}$ is called an *Italian dominating function* (IDF) if every vertex v with $r_i(v) = 0$ is either adjacent to a vertex u with $r_i(u) = 2$ or adjacent to at least two vertices x, y with $r_i(x) = r_i(y) = 1$. Let $\sum_{v \in V} r_i(v)$ be the weight of function r_i . The *minimum Italian dominating function problem* (MinIDF) is to find a minimum weight IDF for G .

The concept of IDF is an important variant of the *Roman dominating function* (RDF) [5], which is motivated by security concerns of the Roman Empire. Such a concern also exists in a wireless sensor network (WSN). Suppose we are installing sensors to monitor a network. Two sensors installed at a node can provide a sufficient monitoring effect for all the locations in its closed neighborhood. One sensor installed at a node can provide a sufficient monitoring effect for itself, and partially monitor its neighboring locations. Two partially monitoring will be sufficient. We have to decide on an installation strategy such that the whole network can be sufficiently monitored using the minimum number of sensors. This is a MinIDF problem.

There are many graph theoretical studies on the MinIDF problem. It is also proved to be NP-hard [2]. But we have not found any study on MinIDF from the approximation algorithm point of view.

In some applications, it is desirable that the monitoring sensors can share information among themselves, where sharing information is guaranteed by connectivity. Motivated by such a consideration, we propose the *minimum connected Italian dominating function problem* (MinCIDF). Call a function $r_{ci} : V(G) \rightarrow \{0, 1, 2\}$ a *connected Italian dominating function* (CIDF) if it is an IDF of G and the subgraph of G induced by $\{v \in V \mid r_{ci}(v) = 1 \text{ or } r_{ci}(v) = 2\}$ is connected. The goal of the MinCIDF problem is to compute a minimum weight CIDF of G .

In this paper, we propose approximation algorithms for MinIDF and MinCIDF in a general graph, with theoretically guaranteed approximation ratios.

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1.1. Related works

Motivated by Stewart’s article titled “Defend the Roman Empire” [17], Cockayne et al. [5] formulated the concept of *Roman dominating function* (RDF), which asks every vertex with zero value to be adjacent with a vertex with value 2. Finding such a function with the minimum weight is known as the *minimum Roman dominating function problem* (MinRDF). RDF has many variants, among which IDF is an important one, which was proposed by Chellali et al. in [2] and is also called Roman- $\{2\}$ -domination.

There are extensive studies on RDF and its variants. For comprehensive studies on RDF, see the monograph [3] and the references therein.

Most studies on IDF are from the graph theoretical point of view [1,7,8,10,11,15,18], while algorithmic studies are rare. It was proved by Chellali et al. [2] that the decision problem for MinIDF is NP-complete, even for bipartite graphs, and Poureidi et al. [14] proved its NP-complete for planar graphs. For special graphs with simple structures, polynomial time algorithms exist. Poureidi et al. [14] and Chen et al. [4] proposed linear-time algorithms for computing the Italian domination number of unicyclic graphs and block graphs, respectively.

Since the MinIDF problem is NP-hard in general, we are interested in designing approximation algorithms. Even for MinRDF, there are only a few studies on how to compute an approximate solution in a general graph. Making use of an approximation algorithm for the minimum dominating set problem, the MinRDF problem admits a $2(1 + \ln(\Delta + 1))$ -approximation [12]. This ratio can be improved to $1 + \ln(\Delta + 1)$ by transforming the MinRDF problem into a minimum submodular cover problem [9]. Li et al. [9] also presented a greedy algorithm for the *connected Roman dominating function problem* (MinCRDF), achieving approximation ratio at most $(1 + \varepsilon)\ln \Delta + O(1)$ for any $\varepsilon > 0$. For unit disk graphs, Shang et al. [16] designed a 5-approximation algorithm and a 7.5-approximation algorithm for MinRDF and MinCRDF, respectively.

Very recently, there are some approximation algorithms for some variants of MinIDF [13,19]. Padamutham et al. [13] designed a $2(\ln(\Delta - 0.5) + 1.5)$ -approximation algorithm for the *minimum total Roman $\{2\}$ -dominating function problem*, which requires the subgraph induced by non-zero vertices to have no isolated vertices. Wang et al. [19] presented a $6(k + 2)$ -approximation algorithm for the *minimum connected strong k -Roman dominating set problem* in unit ball graphs.

So far, we have not seen any approximation algorithm for the MinIDF problem in a general graph.

1.2. Our contribution

The contributions of this paper are summarized as follows.

(i) For the MinIDF problem, we present a $(2 + \ln(\frac{1}{2}\Delta + 1))$ -approximation algorithm, where $\Delta > 0$ is the maximum degree of the graph. The algorithm employs a greedy strategy. It is known that for a *minimum submodular cover problem* (MinSubmC) $\min_{f(\mathbf{x}) \geq k} \mathbf{c}^T \mathbf{x}$, in which f , called the *potential function*, is non-negative integral, monotone nondecreasing and submodular, a greedy algorithm can achieve approximation ratio $1 + \ln \gamma$, where $\gamma = \max\{f(v) : v \in V\}$ (see [6]). In fact, the $(1 + \ln(\Delta + 1))$ -approximation for MinRDF was obtained by transforming the MinRDF problem into a MinSubmC problem. Unfortunately, for the MinIDF problem, we failed to find out such a transformation. However, we propose a new operation “ $\oplus$ ”, and construct a function f with non-negative half-integral, “submodular-like” and “monotone-like” properties under $\oplus$. Using such a potential function f to control the progress of the greedy process, we managed to obtain a logarithmic approximation ratio for the MinIDF problem.

(ii) We propose the MinCIDF problem and present a greedy algorithm achieving approximation ratio at most $4 + \ln(\frac{1}{2}\Delta - 1)$. With the connectivity requirement, it is much more difficult to construct a submodular-like and monotone-like potential function. In fact, the potential function F we constructed is neither monotone non-increasing nor monotone non-decreasing, and it is not submodular-like. However, we find that F has a *relaxed submodular-like property*, which successfully leads to a logarithmic approximation.

The rest of the content is arranged as follows. Approximation algorithms for MinIDF and MinCIDF are presented and analyzed in Section 2 and Section 3, respectively. Section 4 concludes the paper and discusses some future research directions.

2. Greedy algorithm for MinIDF

2.1. Preliminaries

Given a graph G with vertex set V and edge set E , the *open neighborhood* $N_G(v)$ of a vertex $v \in V$ contains all vertices adjacent to v , and the *closed neighborhood* of v is the set $N_G[v] = N_G(v) \cup \{v\}$. The *open neighborhood* of a set $C \subseteq V$ is the set $N_G(C) = \bigcup_{v \in C} N_G(v) \setminus C$, and the *closed neighborhood* of C is the set $N_G[C] = N_G(C) \cup C$. Denote by $\Delta = \max\{|N_G(v)| : v \in V(G)\}$ the *maximum degree* of G . We will omit the subscript G for simplicity when it does not cause ambiguity. Given a vertex set $C \subseteq V$, the subgraph $G[C]$ induced by C is the subgraph of G on vertex set C , two vertices $u, v \in C$ are adjacent in $G[C]$ if they are adjacent in G . We say that a vertex set is *connected* if it induces a connected subgraph.

Consider a finite ground set X , a set function $f : 2^X \rightarrow \mathbb{N}$ is *submodular* if and only if $f(A \cup x) - f(A) \geq f(B \cup x) - f(B)$ holds for any subsets $A \subseteq B \subseteq X$ and any element $x \in X \setminus B$. The set function f is *monotone nondecreasing* if $f(A) \leq f(B)$ holds for any subsets $A \subseteq B$. The set function f is *normalized* if $f(\emptyset) = 0$. These concepts play important roles in the performance guarantee of greedy algorithms for submodular optimization problems. But for MinIDF, they are not applicable. We have to extend these concepts to suit the speciality of the problem.

A function $r : V \mapsto \{0, 1, 2\}$ can be identified with a vertex subset $C \subseteq V$ in the following way: $C = C^1 \cup C^2$, where C^1 consists of all those vertices $v \in V$ with $r(v) = 1$ and C^2 consists of all those vertices $v \in V$ with $r(v) = 2$. Note that in notation $C = C^1 \cup C^2$, sets C^1 and C^2 are *disjoint*. In the following, we use r^C to denote the $\{0, 1, 2\}$ -function corresponding to C .

With the above identification in mind, we define an operation $\oplus$ as follows. For two sets $C = C^1 \cup C^2$ and $D = D^1 \cup D^2$, define

$$C \oplus D = (C \oplus D)^1 \cup (C \oplus D)^2,$$

where $(C \oplus D)^1$ is the symmetric difference of C^1 and D^1 , $(C \oplus D)^2 = C^2 \cup D^2 \cup (C^1 \cap D^1)$. Intuitively, the value of v will be enhanced to 2 if it has value 1 under both r^C and r^D . Otherwise, the value of v will be the higher one under r^C and r^D .

Note that a set $C = C^1 \cup C^2$ can be viewed as a multi-set $\mathcal{C}$, in which every $v \in C^1$ appears once in $\mathcal{C}$ and every $v \in C^2$ appears twice in $\mathcal{C}$. Conversely, a multi-set $\mathcal{C}$ in which every element appears at most twice can be viewed as a set $C = C^1 \cup C^2$. With this understanding, $\oplus$ can be extended to operate on multi-sets, for example, $D \oplus \mathcal{C}$ refers to $D \oplus C$. It should be noted that the weight of r^C equals $|\mathcal{C}|$.

For two vertex subsets A, B , we say that $A \subseteq_{\oplus} B$ if $r^A(v) \leq r^B(v)$ holds for any $v \in V$. A set function f (how f works for a set A of the form $A^1 \cup A^2$ will be clear later) is said to be $\oplus$ -*monotone nondecreasing* if $f(A) \leq f(B)$ holds for any $A \subseteq_{\oplus} B$.

For two vertex subsets $A, C \subseteq V$ and a set function f , define the $\oplus$ *marginal profit* of C over A as

$$f_A^{\oplus}(C) = f(A \oplus C) - f(A).$$

In the following, we mainly use $f_A^{\oplus}(v)$, where the value of v is 1 by default (so in $A \oplus \{v\}$, the value of v will be 1 if $v \notin A$, and will be 2 if $v \in A^1$ or A^2). We say that set function f is $\oplus$ -*submodular* if

$$f_A^{\oplus}(v) \geq f_B^{\oplus}(v) \text{ holds for any } A \subseteq_{\oplus} B \text{ and } v \in V.$$

2.2. The algorithm

For a vertex subset $C = C^1 \cup C^2$, a vertex v with $r^C(v) = 0$ is *totally dominated* by C if $\sum_{u \in N(v)} r^C(u) \geq 2$, v is *partially dominated* by C if $\sum_{u \in N(v)} r^C(u) = 1$. Denote by $NT(C)$ and $NP(C)$ the set of totally dominated and partially dominated vertices by C , respectively. Note that both totally dominated and partially dominated refer to vertices of zero values under r^C . A vertex v is said to be *dominated in C* if $v \in C \cup NT(C)$. Clearly, C corresponds to an IDF if and only if all vertices are dominated.

Define the *domination degree* of vertex v with respect to $C = C^1 \cup C^2$ as

$$Ddeg_C(v) = \begin{cases} 1, & \text{if } v \text{ is dominated in } C, \\ 1/2, & \text{if } v \text{ is partially dominated by } C, \\ 0, & \text{otherwise} \end{cases}$$

Defined potential function $f : 2^V \mapsto \mathbb{R}^+$ to be

$$f(C) = \sum_{v \in V} Ddeg_C(v). \quad (1)$$

Intuitively, every dominated vertex contributes 1 to the potential, and every partially dominated vertex contributes 1/2 to the potential. Note that

$$C \text{ corresponds to an IDF if and only if } f(C) = |V|. \quad (2)$$

The greedy algorithm is described in Algorithm 1. Note that a vertex v can be selected more than once by line 3 of the algorithm. The first time it is selected, its value in $C \oplus \{v\}$ will be 1. The second time it is selected, its value in $C \oplus \{v\}$ will

be 2. We shall prove in the next subsection that the algorithm will never select a vertex more than twice, and as long as $f(C) < |V|$, there exists a vertex v with $f_C^\oplus(v) > 0$.

Algorithm 1 Greedy Algorithm for MinIDF

Input: A graph G with vertex set V and edge set E .

Output: An Italian dominating function r_i for G .

1: $C^1, C^2 \leftarrow \emptyset, C \leftarrow C^1 \cup C^2$

2: **while** $f(C) < |V|$ **do**

3: $v \leftarrow \arg \max_{v \in V} f_C^\oplus(v)$

4: $C \leftarrow C \oplus \{v\}$

5: **end while**

6: Output the function r_i corresponding to C .

Remark 2.1. It should be remarked that by our convention that the value of v in $f_C^\oplus(v)$ is 1 by default, line 3 of the algorithm indicates that the algorithm can only choose vertices “one by one”, that is, it will not choose a vertex $v \notin C$ and set its value to be 2 directly.

2.3. Performance analysis

The following lemma shows that before its termination, the algorithm can always make a positive progress, and a vertex is selected at most twice by the algorithm.

Lemma 2.2. For any vertex set C with $f(C) < |V|$, there exists a vertex v satisfying $f_C^\oplus(v) > 0$. Furthermore, the algorithm will never select any vertex more than twice.

Proof. Note that $f(C) < |V|$ implies that there exists a vertex $v \notin C$ such that $Ddeg_C(v) < 1$. Adding v into C , we have $Ddeg_{C \oplus v}(v) = 1$. Hence $Ddeg_{C \oplus v}(v) - Ddeg_C(v) > 0$. Since $Ddeg_{C \oplus v}(u) - Ddeg_C(u) \geq 0$ holds for any $u \in V \setminus \{v\}$, the first half of the lemma follows from the definition of f in (1). Note that if v already belongs to C^2 , then $f_C^\oplus(v)$ will be zero. The second half of the lemma follows. $\square$

The next lemma shows that the potential function f defined in the above subsection has “submodular-like” and “monotone-like” behaviors.

Lemma 2.3. The function f defined in the above subsection satisfies:

(i) $f(\emptyset) = 0$;

(ii) f is $\oplus$ -monotone nondecreasing;

(ii) f is $\oplus$ -submodular.

Proof. Property (i) is obvious.

For property (ii), note that $A \subseteq_\oplus B$ implies that the domination degree $Ddeg_A(v) \leq Ddeg_B(v)$ for any vertex $v \in V$, and thus $f(A) \leq f(B)$.

For property (iii), note that for a vertex set C and a vertex $v \notin C^2$,

$$Ddeg_{C \oplus \{v\}}(u) - Ddeg_C(u) = \begin{cases} 0, & \text{if } u \text{ is dominated in } C \text{ or } u \notin N[v], \\ 1/2, & \text{otherwise.} \end{cases}$$

Consider two vertex subsets A, B with $A \subseteq_\oplus B$. Note that if u is dominated in A , then u is also dominated in B . Hence for any vertex $v \notin B^2$, we have $Ddeg_{A \oplus \{v\}}(u) - Ddeg_A(u) \geq Ddeg_{B \oplus \{v\}}(u) - Ddeg_B(u)$ for any $u \in V$, and thus $f_A^\oplus(v) \geq f_B^\oplus(v)$. While for $v \in B^2$, we have $B \oplus \{v\} = B$, and thus $f_B^\oplus(v) = 0$. In this case, $f_A^\oplus(v) \geq f_B^\oplus(v)$ is also true. $\square$

Now we analyze the approximate ratio of the algorithm.

Theorem 2.4. Algorithm 1 is a $(2 + \ln \gamma)$ -approximation algorithm for MinIDF, where $\gamma = \max\{f(v) \mid v \in V\} = \frac{1}{2}\Delta + 1$ and Δ is the maximum degree of the graph.

Proof. Suppose Algorithm 1 chooses $v_1, \dots, v_g$ in order. Denote $C_i = \{v_1\} \oplus \dots \oplus \{v_i\}$ for $i \in \{1, \dots, g\}$. Note that the weight of the final output r_i equals $|C_g|$, where C_i is the multi-set corresponding to C_i .

Let r_i^* be a minimum weight IDF and $Y = \{v \in V \mid r_i^*(v) = 1 \text{ or } r_i^*(v) = 2\}$. Let $\mathcal{Y}$ be the multi-set corresponding to set Y . Then $opt = \sum_{v \in Y} r_i^*(v) = |\mathcal{Y}|$ is the optimal value. Write $\mathcal{Y} = \{y_1, y_2, \dots, y_{opt}\}$, $\mathcal{Y}_0 = \emptyset$, and $\mathcal{Y}_j = \{y_1, \dots, y_j\}$ for $j = 1, \dots, opt$. By the greedy criterion, we have

$$f(C_{i+1}) - f(C_i) = f_{C_i}^\oplus(v_{i+1}) \geq f_{C_i}^\oplus(y_j) \text{ holds for any } j \in \{1, \dots, opt\}.$$

Then by Lemma 2.3, we have

$$\begin{aligned} f(C_{i+1}) - f(C_i) &\geq \frac{1}{opt} \sum_{j=1}^{opt} f_{C_i}^{\oplus}(y_j) \geq \frac{1}{opt} \sum_{j=1}^{opt} f_{C_i \oplus \mathcal{Y}_{j-1}}^{\oplus}(y_j) \\ &= \frac{1}{opt} (f(C_i \oplus \mathcal{Y}_{opt}) - f(C_i)) = \frac{1}{opt} (|V| - f(C_i)), \end{aligned}$$

where the last equality uses $|V| \geq f(C_i \oplus \mathcal{Y}_{opt}) \geq f(Y) = |V|$. It follows that

$$|V| - f(C_{i+1}) \leq (|V| - f(C_i)) \cdot \left(1 - \frac{1}{opt}\right).$$

Denote $a_i = |V| - f(C_i)$. Recursively using this relation and because $a_0 = |V|$, we have

$$a_i \leq a_0 \left(1 - \frac{1}{opt}\right)^i \leq |V| \cdot e^{-i/opt}, \quad (3)$$

where the second inequality uses $1 + x \leq e^x$ (for any $x \in \mathbb{R}$).

For the function $r'_i(v) = 1$ ($\forall v \in V$), it is easy to verify that r'_i is an IDF. So, $0 < opt \leq \sum_{v \in V} r'_i(v) = |V|$. By the $\oplus$ -monotonicity of f , we see that a_i monotonely decreases from $|V|$ to 0. So, there is an index $i_0 \in \{1, \dots, g\}$ such that $a_{i_0+1} < opt \leq a_{i_0}$. Using (3) for i_0 , we have

$$opt \leq a_{i_0} \leq |V| \cdot e^{-i_0/opt},$$

which implies

$$i_0 \leq opt \cdot \ln \frac{|V|}{opt}. \quad (4)$$

Note that each iteration increases the f -value by at least $1/2$. So, having obtained C_{i_0+1} , the number of iterations needed to bring a_{i_0+1} down to 0 is at most $2(opt - 1/2) = 2opt - 1$ (note that $a_{i_0+1} < opt$ implies $a_{i_0+1} \leq opt - 1/2$). Since each iteration adds exactly one vertex into the multi-set (see Remark 2.1), we have

$$|C_g| \leq |C_{i_0+1}| + 2opt - 1.$$

Combining this with (4), the weight of the final output r_i is

$$g = |C_g| \leq (i_0 + 1) + (2opt - 1) \leq opt \left(\ln \frac{|V|}{opt} + 2 \right).$$

Using the $\oplus$ -submodularity of f , we have $|V| = f(Y) = f_{\emptyset}^{\oplus}(\mathcal{Y}) = \sum_{j=1}^{opt} f_{\mathcal{Y}_{j-1}}^{\oplus}(y_j) \leq \sum_{j=1}^{opt} f_{\emptyset}^{\oplus}(y_j) = \sum_{j=1}^{opt} f(y_j) \leq opt \cdot \gamma$. Hence $g \leq opt (\ln \gamma + 2)$. Note that for any vertex v , we have $f(v) = 1 + \deg(v)/2$ where $\deg(v)$ is the degree of v . So, $\gamma = 1 + \Delta/2$. The approximation ratio is proved. $\square$

3. Greedy algorithm for the MinCIDF

In this section, we assume that $G = (V, E)$ is a connected graph with $|V| \geq 3$.

The greedy algorithm depends on a potential function F defined as

$$F(C) = p(C) + q(C),$$

where $p(C)$ and $q(C)$ are defined as follows. For a vertex set C , let $G[C]$ be the subgraph of G induced by C , and $p(C)$ be the number of connected components of $G[C]$. For $C = C^1 \cup C^2$, let $E_N(C)$ be the set of edges with one end in $NT(C)$ and another end in C . The graph $G|_{(C)}$ has vertex set V and edge set $E(G[C]) \cup E_N(C)$. Denote by $\mathcal{B}_C$ the set of connected components of $G|_{(C)}$. There are three types of connected components in $\mathcal{B}_C$: a component containing some vertex in C , a vertex in $V \setminus C$ which is not adjacent with C , and a vertex in $NP(C)$. Call them type 1, type 2, type 3 components, respectively. Define the *connectivity degree* of a connected component $B \in \mathcal{B}_C$ as

$$Cdeg_C(B) = \begin{cases} 1/2, & \text{if } B \text{ is a type 3 component of } G|_{(C)}, \\ 1, & \text{otherwise.} \end{cases}$$

Define $q(C) = \sum_{B \in \mathcal{B}_C} Cdeg_C(B)$.

The following lemma shows that function F can be used to judge whether a vertex set C corresponds to a CIDF.

Lemma 3.1. A vertex set $C \subseteq V$ corresponds to a CIDF of G if and only if $F(C) = 2$.

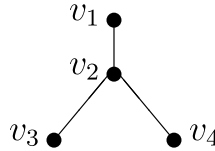


Fig. 1. An example showing that F is not monotone and not submodular.

Proof. If C is a CIDE, then $G[C]$ is connected. So $p(C) = 1$. Since every vertex $v \in V \setminus C$ belongs to $NT(C)$, it is linked to $G[C]$ by some edge in $E_N(C)$. So, $G_{|C}$ is connected, and thus $q(C) = 1$. Hence $F(C) = 2$.

Conversely, suppose $F(C) = 2$. Note that $C \neq \emptyset$ because $F(\emptyset) = |V|$ and we have assumed $|V| \geq 3$. So, $p(C) \geq 1$ and $q(C) \geq 1$. In order that $F(C) = 2$, we must have $p(C) = q(C) = 1$. Then, both $G[C]$ and $G_{|C}$ are connected (if $G_{|C}$ has at least two connected components, then $q(C) \geq 1.5$), which implies that C is a CIDE. $\square$

Remark 3.2. Note that F is neither monotone non-increasing nor monotone non-decreasing. For example, in Fig. 1, we have $F(\{v_1\}) = 4.5 > 4 = F(\emptyset) > 3.5 = F(\{v_2\})$. Also note that F is neither $\oplus$ -submodular nor $\oplus$ -supermodular (a function f is $\oplus$ -supermodular if $-f$ is $\oplus$ -submodular). For example, in Fig. 1, we have $F_{\emptyset}^{\oplus}(v_1) = 0.5 > 0 = F_{\{v_2, v_2\}}^{\oplus}(v_1) > -0.5 = F_{\{v_2\}}^{\oplus}(v_1)$. So, the method in the previous section cannot be used here. We have to explore more structural properties of the MinCIDE problem.

The greedy algorithm for MinCIDE is described in Algorithm 2. It is similar to that of Algorithm 1, using potential function $-F$ instead of f . For similar reasons as in the previous section, the algorithm chooses vertices one by one, and each vertex can be chosen at most twice.

Algorithm 2 Greedy Algorithm for MinCIDE

Input: A graph G with vertex set V and edge set E .

Output: A connected Italian dominating function r_{ci} for G .

1: $C^1, C^2 \leftarrow \emptyset, C = C^1 \cup C^2$

2: **while** $F(C) > 2$ **do**

3: $v \leftarrow \arg \max_{v \in V} -F_C^{\oplus}(v)$

4: $C \leftarrow C \oplus v$

5: **end while**

6: Output the function r_{ci} corresponding to C .

To facilitate the analysis of performance, we first formulate the expressions of $-p_C^{\oplus}(v)$ and $-q_C^{\oplus}(v)$ for any vertex set $C \subseteq V$ and any vertex $v \in V$.

Denote by $NC_C(v)$ the set of connected components of $G[C]$ adjacent to v (the connected component of $G[C]$ containing v is also regarded as belonging to $NC_C(v)$). Note that adding v into C , those connected components of $NC_C(v)$ are merged into one big connected component of $G[C \oplus \{v\}]$. So, we have the following observation.

Observation 3.3. For any vertex set C and any vertex v , $-p_C^{\oplus}(v) = |NC_C(v)| - 1$. As a consequence, $-p_C^{\oplus}(v) \geq -1$, equality holds if and only if $NC_C(v) = \emptyset$, that is, $v \notin C$ and v is not adjacent with any vertex of C .

Note that adding v into C refers to $C \oplus \{v\}$. Observation 3.3 holds for any $v \in V$. In particular, if $v \in C$, then $-p_C^{\oplus}(v) = 0$.

To formulate an expression for $-q_C^{\oplus}(v)$, we consider $\mathcal{B}_C$, the set of connected components of $G_{|C}$. Let $B_C(v)$ be the component in $\mathcal{B}_C$ containing v . For $i \in \{1, 2, 3\}$, denote by $NB_C^i(v)$ the set of type i components of $\mathcal{B}_C \setminus \{B_C(v)\}$ adjacent with v . Let $NNB_C(v)$ denote the set of type 1 components of $\mathcal{B}_C \setminus (\{B_C(v)\} \cup NB_C^1(v))$ adjacent with some vertex of $NB_C^3(v)$. It should be emphasized that the sets $NB_C^i(v)$ and $NNB_C(v)$ exclude $B_C(v)$, otherwise, some components might be repeatedly counted in the following expression. Consider the example in Fig. 2 (a). $B(v) = \{v\}$ is a type 3 component of $G_{|C}$, $NB^1(v) = \{B_1, B_2\}$, $NB^2(v) = \{v_2\}$, $NB^3(v) = \{v_1, v_3, v_4\}$, and $NNB_C(v) = \{B_3\}$. Note that although B_2 is a type 1 component associated with the type 3 component v_1 , it does not belong to $NNB_C(v)$ because it belongs to $NB_C^1(v)$.

Observation 3.4. For any vertex set C and any vertex v ,

$$-q_C^{\oplus}(v) = |NB_C^1(v)| + \frac{1}{2}|NB_C^2(v) \cup NB_C^3(v)| + |NNB_C(v)| + Cdeg_C(B_C(v)) - 1.$$

As a consequence of this expression, we always have $-q_C^{\oplus}(v) \geq 0$.

The observation can be illustrated by the example in Fig. 2 (a). Adding v into C , all those components in $NB_C^1(v)$ are merged into the component $B_C(v)$, decreasing the q -value by the amount of $|NB_C^1(v)|$. A type 2 component in $NB_C^2(v)$

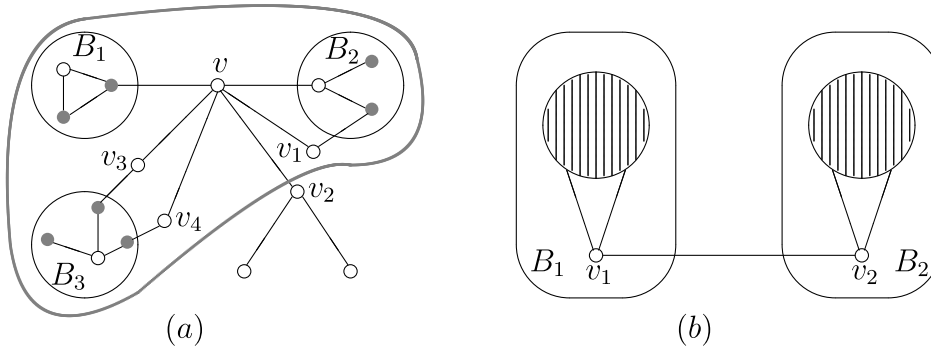


Fig. 2. (a) An illustration for the expression in [Observation 3.4](#). Gray vertices belong to C . Light circles indicate components of $G|_{(C)}$. The blackened curve marks the component of $G|_{(C \oplus \{v\})}$ which is the expansion of $B(v)$. (b) Illustration for Case 2 in the proof of [Lemma 3.5](#).

becomes a type 3 component of $G|_{(C \oplus \{v\})}$, whose contribution to the q -value is decreased from 1 to $1/2$. So, the total contribution of those components in $NB_C^2(v)$ to the q -value is decreased by $\frac{1}{2}|NB_C^2(v)|$. A type 3 component in $NB_C^3(v)$, which contains a single vertex, say u , becomes a totally dominated vertex in $C \oplus \{v\}$, and thus $\{u\}$, together with the type 1 component adjacent with u (if it does not belong to $NB_C^1(v)$, the set of components that are already merged into $B(v)$), are merged into $B(v)$. So, the total contribution incurred by those components in $NB_C^3(v)$ to the q -value is decreased by $\frac{1}{2}|NB_C^3(v)| + |NNB_C(v)|$. As to $B(v)$, its contribution to $q(C)$ is $Cdeg_C(B_C(v))$, while in $C \oplus \{v\}$, it is expanded to a component with $Cdeg_{C \oplus \{v\}}$ -value 1, resulting in a decrease of $Cdeg_C(B_C(v)) - 1$.

Although [Fig. 2](#) only illustrates the observation by a type 3 component $\{v\}$, the observation is valid for any $v \in V$.

Note that the expression of $-q_C^\oplus(v)$ implies $-q_C^\oplus(v) \geq -1/2$, equality holds only when $NB_C^1(v) = NB_C^2(v) = NB_C^3(v) = NNB_C(v) = \emptyset$ and $Cdeg_C(v) = 1/2$. By definition, $Cdeg_C(v) = 1/2$ implies that $\{v\}$ is a type 3 component in $G|_{(C)}$, and thus v is adjacent to a vertex in C^1 . But this implies that v is adjacent with a type 1 component of $G|_{(C)}$, contradicting $NB_C^1(v) = \emptyset$. Then $-q_C^\oplus(v) \geq 0$ follows from the observation that the value of q is half integral.

The following lemma shows that the algorithm can always make a positive progress before the F -value is decreased to 2, and thus at the termination, the function r_{ci} output by [Algorithm 2](#) is indeed a CIDF.

Lemma 3.5. For any vertex set $C = C^1 \cup C^2$ with $F(C) > 2$, there exists a vertex $v \in V$ such that $-F_C^\oplus(v) > 0$.

Proof. By [Lemma 3.1](#), C does not corresponds to a CIDF. We distinguish two cases. **Case 1.** C does not correspond to an IDF.

In this case, $C \cup NT(C) \neq V$. Because G is connected, there is a vertex $v \in V \setminus (C \cup NT(C))$ which is adjacent with a vertex $u \in C \cup NT(C)$.

If $u \in C$, since $v \notin C \cup NT(C)$, we have $v \in NP(C)$. Hence $\{v\}$ is a type 3 component of $G|_{(C)}$, $|NB_C^1(v)| \geq 1$ and $Cdeg_C(v) = 1/2$. Adding v into C^1 , by [Observation 3.4](#), we have $-q_C^\oplus(v) \geq 1/2$. Furthermore, since $NC_C(v) \neq \emptyset$, by [Observation 3.3](#), we have $-p_C^\oplus(v) \geq 0$. Hence $-F_C^\oplus(v) = -p_C^\oplus(v) - q_C^\oplus(v) > 0$.

If v is not adjacent with any vertex in C , then $\{v\}$ is a type 2 component of $G|_{(C)}$, and $u \in NT(C)$. Adding u into C^1 , since $NC_C(u) \neq \emptyset$, we have $-p_C^\oplus(u) \geq 0$. The existence of $\{v\}$ implies that $|NB_C^2(u)| \geq 1$. Note that $Cdeg_C(B(u)) = 1$. Hence [Observation 3.4](#) leads to $-q_C^\oplus(u) \geq 1/2$. In this case, we have $-F_C^\oplus(u) > 0$.

Case 2. C corresponds to an IDF, but $G[C]$ is not connected.

If there is a vertex $v \in V \setminus C$ with $|NC_C(v)| \geq 2$ (that is, if some connected components of $G[C]$ have a common neighbor v), then $-p_C^\oplus(v) \geq 1$. Since we always have $-q_C^\oplus(v) \geq 0$, in this case, we have $-F_C^\oplus(v) > 0$.

Next, suppose no connected components of $G[C]$ have a common neighbor. The proof can be illustrated by [Fig. 2 \(b\)](#). Because C corresponds to an IDF, every vertex is dominated, and thus all connected components of $G|_{(C)}$ are of type 1. Consider two *closest* connected components of $G|_{(C)}$, say B_1 and B_2 . Since they are “closest”, there is an edge v_1v_2 linking them (say $v_1 \in B_1$ and $v_2 \in B_2$). Under the assumption that no connected components of $G[C]$ have a common neighbor, we have $v_1, v_2 \in NT(C)$. Note that $B_1 = B(v_1)$ has $Cdeg_C(B_1) = 1$, and the existence of B_2 implies $|NB_C^1(v_1)| \geq 1$. Hence adding v_1 into C^1 leads to $-q_C^\oplus(v_1) \geq 1$. Since v_1 is dominated by some vertices in C , we have $-p_C^\oplus(v_1) \geq 0$. In any case, we have $-F_C^\oplus(v_1) > 0$. $\square$

To analyze the approximation ratio, similar notations as in the previous section are used. Suppose [Algorithm 2](#) selects $v_1, \dots, v_g$ in order. Denote $C_i = v_1 \oplus \dots \oplus v_i$ for $i = 1, \dots, g$ and $C_0 = \emptyset$. Let C_i be the multi-set corresponding to C_i . Note that $|C_g|$ is the weight of the final output function r_{ci} .

Let r_{ci}^* be a minimum weight CIDF. Denote $Y = \{v \mid r_{ci}^*(v) = 1 \text{ or } r_{ci}^*(v) = 2\}$. Different from the previous section, ordering elements in Y plays an important role in the performance analysis. We order the elements in Y such that every

vertex (except for the first one) is adjacent with some vertex ordered before it. Since $G[Y]$ is connected, such an ordering exists. The multi-set $\mathcal{Y}$ corresponding to set Y is ordered similarly, with those copies of a same vertex $v \in Y^2$ put together. Write $\mathcal{Y} = \{y_1, \dots, y_{opt}\}$. Denote $\mathcal{Y}_0 = \emptyset$ and $\mathcal{Y}_j = \{y_1, \dots, y_j\}$ for $j = 1, \dots, opt$. By the above ordering method, we see that the induced subgraph $G[\mathcal{Y}_j]$ is connected for $j = 1, \dots, opt$. Note that $opt = \sum_{v \in Y} r_{cl}^*(v) = |\mathcal{Y}|$ is the optimal value of the MinCIDEF problem.

The following lemma plays an important role in obtaining desired approximation ratio.

Lemma 3.6. For any $j \in \{1, \dots, opt\}$ and $i \in \{1, \dots, g\}$,

$$-F_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^\oplus(y_j) \leq -F_{C_{i-1}}^\oplus(y_j) + 1. \quad (5)$$

Proof. This is obvious for $j = 1$. Hence in the following, we assume $j \geq 2$.

By the expression $-p_C^\oplus(v) = |NC_C(v)| - 1$ in [Observation 3.3](#), since $G[\mathcal{Y}_{j-1}]$ is connected, $NC_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$ has at most one more connected component than $NC_{C_{i-1}}(y_j)$, namely the one containing $\mathcal{Y}_{j-1}$. So,

$$-p_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^\oplus(y_j) \leq -p_{C_{i-1}}^\oplus(y_j) + 1. \quad (6)$$

Note that equality holds only when $\mathcal{Y}_{j-1}$ is neither adjacent with nor shares a common vertex with any connected component of $NC_{C_{i-1}}(y_j)$.

With the expression of $-q_C^\oplus(v)$ in mind ([Observation 3.4](#)), in the following, we estimate the relation between corresponding terms of $-q_{C_{i-1}}^\oplus(y_j)$ and $-q_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^\oplus(y_j)$.

(1) **NB¹**: Since $G[\mathcal{Y}_{j-1}]$ is connected, we have

$$|NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^1(y_j)| \leq |NB_{C_{i-1}}^1(y_j)| + 1. \quad (7)$$

The “+1” in the right side of (7) will diminish under the following two situations:

(a) $\mathcal{Y}_{j-1}$ is adjacent with or shares a common vertex with some component of $NB_{C_{i-1}}^1(y_j)$. This is because “+1” in the right side of (7) comes from the type 1 component of $G_{|C_{i-1} \oplus \mathcal{Y}_{j-1}|}$ that contains $\mathcal{Y}_{j-1}$. If equality holds in (7), then this component must be disjoint from any component of $NB_{C_{i-1}}^1(y_j)$, which contradicts condition (a).

(b) y_j is a vertex of C_{i-1} or is adjacent with some vertex in C_{i-1} . In this case, y_j is a dominated vertex in $C_{i-1} \oplus \mathcal{Y}_{j-1}$, and thus $G[\mathcal{Y}_{j-1}]$, being a connected subgraph adjacent with y_j , is contained in the component $B_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$, the component of $G_{|C_{i-1} \oplus \mathcal{Y}_{j-1}|}$ containing y_j . Since the definition of $NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^1(y_j)$ excludes $B_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$, so $\oplus \mathcal{Y}_{j-1}$ will not incur “+1” in the right side.

Furthermore, under the following situation (c), inequality (7) can be further improved into $|NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^1(y_j)| \leq |NB_{C_{i-1}}^1(y_j)| - 1$:

(c) $\{y_j\}$ is a type 3 component of $G_{|C_{i-1}|}$. In fact, (c) is a special situation of (b), and thus we do not have “+1” due to the connected subgraph $G[\mathcal{Y}_{j-1}]$. Let v be the unique vertex in C_{i-1}^1 that is adjacent with y_j , and let $B(v)$ be the type 1 component of $G_{|C_{i-1}|}$ containing v . Then $B(v) \in NB_{C_{i-1}}^1(y_j)$. For the same reason as in (a), y_j is a dominated vertex in $C_{i-1} \oplus \mathcal{Y}_{j-1}$ and thus $B(v)$ is contained in $B_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$, which is not counted in $NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^1(y_j)$. The disappearance of $B(v)$ from NB^1 results in a further “−1” in the right side of (7).

(2) **NB² $\cup$ NB³**: Note that

$$|NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^2(y_j) \cup NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^3(y_j)| \leq |NB_{C_{i-1}}^2(y_j) \cup NB_{C_{i-1}}^3(y_j)|. \quad (8)$$

This is because adding more vertices into C_{i-1} , the number of type 2 components will not increase. Although some type 3 component, say $\{v\}$, may be created after adding $\mathcal{Y}_{j-1}$ into C_{i-1} , this happens only when $\{v\}$ is a type 2 component of $G_{|C_{i-1}|}$ and v is adjacent with a unique vertex of $\mathcal{Y}_{j-1}$. In such a case, the number of increased type 3 components is balanced by the number of decreased number of type 2 components.

(3) **NNB**: Note that any component in $NNB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$ is associated with a type 3 component of $NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^3(y_j)$. A consequence of the above argument is: for every type 3 component $\{v\}$ in $NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^3(y_j) \setminus NB_{C_{i-1}}^3(y_j)$ (that is, $\{v\}$ is a newly created type 3 component because of adding $\mathcal{Y}_{j-1}$ into C_{i-1}), the unique vertex of $(C_{i-1} \oplus \mathcal{Y}_{j-1})^1$ that is adjacent with v belongs to $\mathcal{Y}_{j-1}$. Let $B(v)$ be the type 1 component of $G_{|C_{i-1} \oplus \mathcal{Y}_{j-1}|}$ adjacent with v . Then $B(v)$ contains $\mathcal{Y}_{j-1}$. Since y_j is adjacent with $\mathcal{Y}_{j-1}$, $B(v)$ is either a type 1 component of $NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^1(y_j)$ or the component $B_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$, in both cases, $B(v)$ is not counted in $NNB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$. Hence

$$|NNB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)| \leq |NNB_{C_{i-1}}(y_j)|. \quad (9)$$

(4) **Cdeg**: By the definition of $Cdeg$, we have

$$Cdeg_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(B_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)) \leq Cdeg_{C_{i-1}}(B_{C_{i-1}}(y_j)) + 1/2, \quad (10)$$

equality is possible only when $Cdeg_{C_{i-1}}(B_{C_{i-1}}(y_j)) = 1/2$, that is, $\{y_j\}$ is a type 3 component in $G|_{C_{i-1}}$.

By the above inequalities, we already have $-F_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^{\oplus}(y_j) \leq -F_{C_{i-1}}^{\oplus}(y_j) + 5/2$. Next, we refine the constant to 1 by considering two cases.

Case 1. $\mathcal{Y}_{j-1}$ is adjacent with or shares a common vertex with a connected component of $NC_{C_{i-1}}(y_j)$.

In this case, condition (a) is also satisfied. So, “+1” in both (6) and (7) diminish. Summing the above inequalities (6) to (10) (ignoring the diminishing terms), we have $-F_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^{\oplus}(y_j) \leq -F_{C_{i-1}}^{\oplus}(y_j) + 1/2$.

Case 2. $\mathcal{Y}_{j-1}$ is neither adjacent with nor shares a common vertex with a connected component of $NC_{C_{i-1}}(y_j)$.

If equality holds in (10), then condition (c) is satisfied, and by the improved inequality $|NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^1(y_j)| \leq |NB_{C_{i-1}}^1(y_j)| - 1$, we have $-F_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^{\oplus}(y_j) \leq -F_{C_{i-1}}^{\oplus}(y_j) + 1/2$ in this case. So in the following, assume that “+1/2” does not appear in (10). Under this assumption, if “+1” in (7) diminishes, then the desired inequality (5) follows. Hence in the following, we also assume that situations (a) and (b) do not occur.

Under these assumptions, every vertex in $\mathcal{Y}_{j-1}$ is a type 2 component of $G|_{C_{i-1}}$. By the ordering method of $\mathcal{Y}$, vertex y_j is adjacent with some vertex, say y_ℓ , of $\mathcal{Y}_{j-1}$ (since we have assumed $j \geq 2$, we have $\mathcal{Y}_{j-1} \neq \emptyset$). If y_j is not a copy of y_ℓ , then $\{y_\ell\}$ is a type 2 component of $NB_{C_{i-1}}^2(y_j)$, which in neither a type 2 component nor a type 3 component of $G|_{C_{i-1} \oplus \mathcal{Y}_{j-1}}$. In this case, (8) is improved into

$$|NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^2(y_j) \cup NB_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^3(y_j)| \leq |NB_{C_{i-1}}^2(y_j) \cup NB_{C_{i-1}}^3(y_j)| - 1. \quad (11)$$

Inequality (11) is also true if $j \geq 3$ and y_j is a copy of y_ℓ (because $G[\mathcal{Y}_{j-1}]$ is connected, a neighbor of y_ℓ in $\mathcal{Y}_{j-1}$ plays the same role as in the above). If y_j is a dominated vertex in $C_{i-1} \oplus \mathcal{Y}_{j-1}$, then $G[\mathcal{Y}_{j-1}]$ is contained in $B_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j)$, and thus “+1” in (7) diminishes. In this case, inequality (5) holds. Hence we assume that y_j is only adjacent with one vertex in $(\mathcal{Y}_{j-1})^1$. Then y_j is a type 3 component in $G|_{C_{i-1} \oplus \mathcal{Y}_{j-1}}$, with

$$Cdeg_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_j) = 1/2 = 1 - 1/2 = Cdeg_{C_{i-1}}(y_j) - 1/2. \quad (12)$$

Combining the expression of $-q_C^{\oplus}$ and (6), (7), (9), (11), (12), we can obtain inequality (5) is also true in this case.

Finally, we are left with the case when $j = 2$ and y_2 is a copy of y_1 . In this case, $G[\mathcal{Y}_1]$ is contained in $B_{C_{i-1} \oplus \mathcal{Y}_{j-1}}(y_2)$ and thus “+1” in (7) diminishes, and thus also leads to the validity of inequality (5). $\square$

The following lemma gives a relationship between $F(C_i)$ and $F(C_{i-1})$.

Lemma 3.7. For $i = 1, 2, \dots, g$,

$$F(C_i) \leq F(C_{i-1}) - \frac{F(C_{i-1}) - 2}{opt} + 1.$$

Proof. By Lemma 3.6, we have

$$-F_{C_{i-1}}^{\oplus}(\mathcal{Y}) = \sum_{j=1}^{opt} (-F_{C_{i-1} \oplus \mathcal{Y}_{j-1}}^{\oplus}(y_j)) \leq \sum_{j=1}^{opt} (-F_{C_{i-1}}^{\oplus}(y_j) + 1).$$

By Lemma 3.1 and the fact that $C_{i-1} \oplus \mathcal{Y}$ corresponds to a CDF of G , we have

$$-F_{C_{i-1}}^{\oplus}(\mathcal{Y}) = F(C_{i-1}) - F(C_{i-1} \oplus \mathcal{Y}) = F(C_{i-1}) - 2.$$

It follows that

$$\sum_{j=1}^{opt} (-F_{C_{i-1}}^{\oplus}(y_j) + 1) \geq F(C_{i-1}) - 2.$$

By the pigeonhole principle, there exists a vertex $y_j \in \mathcal{Y}$ such that

$$-F_{C_{i-1}}^{\oplus}(y_j) + 1 \geq \frac{F(C_{i-1}) - 2}{opt}.$$

Combining this with the greedy strategy of Algorithm 2, we have

$$F(C_{i-1}) - F(C_i) = -F_{C_{i-1}}^{\oplus}(v_i) \geq -F_{C_{i-1}}^{\oplus}(y_j) \geq \frac{F(C_{i-1}) - 2}{opt} - 1.$$

Recollecting the terms, the lemma is proved. $\square$

We are now ready to derive the approximation ratio of Algorithm 2.

Theorem 3.8. Algorithm 2 has approximation ratio at most $4 + \ln(\frac{1}{2}\Delta - 1)$, where Δ is the maximum degree of the input graph.

Proof. Rearrange the terms of the inequality in Lemma 3.7, we have

$$F(C_i) - 2 \leq (F(C_{i-1}) - 2) \left(1 - \frac{1}{opt}\right) + 1. \quad (13)$$

Recursively using inequality (13), we obtain

$$\begin{aligned} F(C_i) - 2 &\leq (F(C_0) - 2) \left(1 - \frac{1}{opt}\right)^i + \sum_{k=0}^{i-1} \left(1 - \frac{1}{opt}\right)^k \\ &= (F(C_0) - 2) \left(1 - \frac{1}{opt}\right)^i + opt \left(1 - \left(1 - \frac{1}{opt}\right)^i\right) \\ &= (F(C_0) - 2 - opt) \left(1 - \frac{1}{opt}\right)^i + opt. \end{aligned}$$

Recollecting the terms, we have

$$F(C_i) - 2 - opt \leq (F(C_0) - 2 - opt) \left(1 - \frac{1}{opt}\right)^i \leq (F(C_0) - 2 - opt) e^{-i/opt}. \quad (14)$$

Lemma 3.5 tells us that every iteration strictly decreases the F -value. Since F takes half-integral values, every iteration decreases the F -value by at least $1/2$. At the termination, $F(C_g) = 2$. Note that each iteration adds exactly one vertex into the multi-set. So, if $F(C_0) \leq 2opt + 2$, then the approximate value, which equals $|C_g|$, is at most $4opt$, and we have approximation ratio at most 4. In the following, we assume that $F(C_0) > 2opt + 2$. Since $F(C_g) = 2 < 2opt + 2$, there is an index i_0 with $F(C_{i_0+1}) < 2opt + 2 \leq F(C_{i_0})$. So, $opt \leq F(C_{i_0}) - 2 - opt$. Combining this with (14) and the fact $F(C_0) = |V|$, we have

$$i_0 \leq opt \cdot \ln \frac{|V| - 2 - opt}{opt}.$$

To decrease the F -value from $F(C_{i_0+1})$ to $F(C_g)$ needs at most $2((2opt + 3/2) - 2) = 4opt - 1$ iterations. Hence

$$|C_g| \leq (4opt - 1) + 1 + i_0 \leq \left(4 + \ln \frac{|V| - 2 - opt}{opt}\right) opt, \quad (15)$$

where the term “+1” corresponds to adding v_{i_0+1} to decrease $F(C_{i_0})$ to $F(C_{i_0+1})$.

Next, we show that

$$|V| - 2 \leq \frac{\Delta}{2} opt. \quad (16)$$

In fact, note that

$$\begin{aligned} -F_{\emptyset}^{\oplus}(v) &= (p(\emptyset) + q(\emptyset)) - (p(v) + q(v)) \\ &= (0 + |V|) - \left(1 + \frac{1}{2}deg(v) + |V| - deg(v)\right) \\ &= \frac{1}{2}deg(v) - 1. \end{aligned}$$

Combining this with Lemma 3.6, we have

$$\begin{aligned} |V| - 2 &= F(\emptyset) - F(Y) = \sum_{j=1}^{opt} \left(-F_{\mathcal{Y}_{j-1}}^{\oplus}(y_j)\right) \leq \sum_{j=1}^{opt} \left(-F_{\emptyset}^{\oplus}(y_j) + 1\right) \\ &\leq opt \cdot \max_{j=1, \dots, opt} \left\{-F_{\emptyset}^{\oplus}(y_j) + 1\right\} \leq \frac{\Delta}{2} opt. \end{aligned}$$

Substituting (16) into (15), the desired approximation ratio is proved. $\square$

4. Conclusions

In this paper, we design a greedy $2 + \ln(\frac{1}{2}\Delta + 1)$ -approximation algorithm and a greedy $4 + \ln(\frac{1}{2}\Delta - 1)$ -approximation algorithm for MinIDF and MinCIDF, respectively. They are the first approximation algorithms for the problems in general

graphs. Submodular-like and monotone-like properties play an important role in the analysis for the MinIDF problem. But for the MinCIDF problem, the potential function is neither monotone-like nor submodular-like. We hope that our method to deal with such difficulties might be useful in studying some other similar problems.

Note that MinIDF is a special *minimum ω -dominating function* (Min ω DF) problem proposed in [10]. Given a graph G , a vector of nonnegative integers $\omega = (\omega_0, \dots, \omega_l)$ with $\omega_0 \geq 1$, and a function $f : V(G) \mapsto \{0, \dots, l\}$, define $V_i = \{v \in V(G) \mid f(v) = i\}$. We say that f is a *ω -dominating function* (ω DF) if for every $i \in \{0, \dots, l\}$ and every vertex $v \in V_i$, $\sum_{u \in N(v)} f(u) \geq \omega_i$. The weight of f is defined to be $\sum_{i=1}^l \omega_i |V_i|$. The Min ω DF problem asks for a minimum weight ω DF of G . Note that the MinIDF problem is a special Min ω DF problem with $\omega = (2, 0, 0)$. Our method can be extended to deal with the Min ω DF problem, by extending the concept of domination degree. However, how to deal with a connected version of the Min ω DF problem is very complicated. Trying to obtain a good approximation algorithm for the connected Min ω DF problem needs more insights and exploration.

Data availability

No data was used for the research described in the article.

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References

- [1] C. van Bommel, Italian domination of cartesian products of directed cycles, *Discrete Appl. Math.* 299 (2021) 82–86.
- [2] M. Chellali, T.W. Haynes, S.T. Hedetniemi, A.A. McRae, Roman {2}-domination, *Discrete Appl. Math.* 204 (2016) 22–28.
- [3] M. Chellali, N.J. Rad, S.M. Sheikholeslami, L. Volkmann, Roman domination in graphs, in: T.W. Haynes, et al. (Eds.), *Topics in Domination in Graphs*, Springer Nature, Switzerland AG, 2020.
- [4] H. Chen, C. Lu, Roman {2}-domination problem in graphs, *Discuss. Math. Graph Theory* 42 (2) (2022) 641–660.
- [5] E.J. Cockayne, P.A. Dreyer, S.M. Hedetniemi, S.T. Hedetniemi, Roman domination in graphs, *Discret. Math.* 278 (1–3) (2000) 11–22.
- [6] D.-Z. Du, K.-I. Ko, X. Hu, *Design and Analysis of Approximation Algorithms*, Springer, 2012.
- [7] M.A. Henning, W.F. Klostermeyer, Italian domination in trees, *Discrete Appl. Math.* 217 (2017) 557–564.
- [8] R. Hernández-Ortiz, L.P. Montejano, J.A. Rodríguez-Velázquez, Italian domination in rooted product graphs, *Bull. Malays. Math. Sci. Soc.* 44 (1) (2021) 497–508.
- [9] K. Li, Y.L. Ran, Z. Zhang, D.-Z. Du, Nearly tight approximation algorithm for (connected) Roman dominating set, *Optim. Lett.* 16 (2022) 2261–2276.
- [10] A.C. Martinez, A. Estrada-Moreno, J.A. Rodríguez-Velázquez, From Italian domination in lexicographic product graphs to ω -domination in graphs, *Ars Math. Contemp.* 22 (1) (2022) P1.04.
- [11] S. Nazari-Moghaddam, M. Chellali, A new upper bound for the perfect Italian domination number of a tree, *Discuss. Math. Graph Theory* 42 (3) (2020) 1005–1022.
- [12] C. Padamutham, V. Palagiri, Algorithmic aspects of Roman domination in graphs, *J. Appl. Math. Comput.* 64 (1–2) (2020) 89–102.
- [13] C. Padamutham, V.S.R. Palagiri, Algorithmic aspects of total Roman {2}-domination in graphs, *Commun. Comb. Optim.* 7 (2) (2022) 183–192.
- [14] A. Poureidi, N.J. Rad, On the algorithmic complexity of Roman {2}-domination (Italian domination), *Iran. J. Sci. Technol. Trans. A Sci.* 44 (3) (2020) 791–799.
- [15] N.J. Rad, M. Hajibaba, On domination, {2}-domination, and Italian domination numbers, *Util. Math.* 111 (111) (2019) 271–280.
- [16] W. Shang, X. Wang, X. Hu, Roman domination and its variants in unit disk graphs, *Discrete Math. Algorithms Appl.* 2 (01) (2010) 99–105.
- [17] I. Stewart, Defend the Roman empire!, *Sci. Am.* 281 (6) (1999) 136–138.
- [18] J. Varghese, V. Anu, S.A. Lakshmanan, Italian domination and perfect Italian domination on Sierpiński graphs, *J. Discrete Math. Sci. Cryptogr.* 24 (7) (2021) 1885–1894.
- [19] L. Wang, Y. Shi, Z. Zhang, Z.-B. Zhang, X. Zhang, Approximation algorithm for a generalized Roman domination problem in unit ball graphs, *J. Comb. Optim.* 39 (2020) 138–148.